Department of Mathematics and Statistics
College of Science
Polytechnic University of the Philippines

Course Code:	MATH 206	**Course Name:**	Linear Algebra and Matrix Theory
Units:	3	**Prerequisite/s:**	MATH 105 or equiv.
Course Description	The course is an introduction to the theory of vectors, matrices and linear transformations. The course covers solvability of linear systems, vectors and matrix operations, determinants, dimension of a space, orthogonality, linear transformations, similarity, eigenvalues and eigenvectors, and diagonalizability. Selected applications are also covered.		

Course Plan

Week	Topic/Activity
1 – 2	Course Orientation
	Vector and Matrix Operations
3 – 5	Linear Systems
	Nonsingular Matrices
	Determinants and Cofactor Expansion
6	Long Exam 1
7 – 9	Subspaces and Linear Independence
	Basis and Dimension
	Orthonormal Bases and Gram-Schmidt Process
10 – 11	Linear Transformations
	Invertible Linear Transformations
12	Long Exam 2
13 – 14	Fundamental Spaces of a Matrix
	The Projection Theorem
15 – 16	Eigenvalues and Eigenvectors
	Diagonalizable Operators
	Spectral Theorem
17 – 18	Long Exam 3
	Final Exam

Grading System
(50 - Base)

Final Grade Components:

- [CS] Class Standing
  - average of long exams
- [ME] Major Exam
  - final exam

Final Grade [FG]:

[FG] = (0.7)[CS] + (0.3)[ME]

NOTE: [FG] is rounded off to the nearest integer and is converted to the University point grade system.

Final Grade	Point Grade
97 – 100	1.00
94 – 96	1.25
91 – 93	1.50
88 – 90	1.75
85 – 87	2.00
82 – 84	2.25
79 – 81	2.50
76 – 78	2.75
75	3.00
50 – 74	5.00
Withdrawn	W
Dropped	D
Incomplete	INC

References

1. Free e-book: **Linear Algebra**, 4th Ed, by Jim Hefferon.
   Download at `https://hefferon.net/linearalgebra/index.html`
2. **Elementary Linear Algebra**, 8th Ed, by Ron Larson

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